Random Variables & Joint Distributions

Discrete and continuous random variables · Joint, marginal and conditional distributions

Random Variable

Concepts

- A random variable is a formal mathematical function that maps the outcome of a random event to a numerical value. In formal notation, it is a function $X: \Omega \to \mathbb{R}$, where $\Omega$is the sample space of all possible outcomes.
- Instead of dealing with abstract events like "it rained" or "the coin landed heads," a random variable assigns a number to these events so they can be processed algebraically
- It converts outcomes into numbers so we can compute with them
- This shift from **sets and events** to **functions and numbers** allows you to calculate expected values, variances, and plug probabilistic behaviors directly into optimization algorithms and mathematical models.
- Range R_X = set of values X can take

Worked example

- Toss 3 fair coins; let X = number of heads
- Ω has 8 equally likely outcomes; X(HHT) = 2, X(TTT) = 0
- P(X=0)=1/8, P(X=1)=3/8, P(X=2)=3/8, P(X=3)=1/8
- Check: 1/8+3/8+3/8+1/8 = 1
- P(X ≥ 2) = 3/8 + 1/8 = 1/2

Type	Definition	Notation	Example
Discrete	Takes on a countable number of distinct values (integers).		
Continuous	Takes on an infinite, uncountable number of values within a range.		
Joint	The probability of two or more random variables occurring simultaneously.		
Marginal	The probability of one variable occurring, calculated by summing/integrating out all other variables in a joint distribution.		
Conditional	The probability of a variable occurring given that the outcome of another variable is already known.		

Discrete Random Variables

Concepts

- PMF: $p(x) = P(X = x)$, with $p(x) \geq 0$ and $\sum p(x) = 1$
- CDF: $F(x) = P(X \leq x) = \sum_{t \leq x} p(t)$, a step function
- $E[X] = \sum x \cdot p(x)$; $Var(X) = E[X^2] - (E[X])^2$
- Common laws: Bernoulli, Binomial(n,p), Geometric(p), Poisson(λ)

Worked example (network)

- 5 packets are sent; each is lost independently with p = 0.1. X = packets lost ~ Binomial(5, 0.1)
- $P(X=2) = C(5,2)(0.1)^2(0.9)^3 = 10 \times 0.01 \times 0.729 = 0.0729$
- $P(X=0) = 0.9^5 = 0.5905$
- $P(X \geq 1) = 1 - 0.5905 = 0.4095$
- $E[X] = 5(0.1) = 0.5$; $Var(X) = 5(0.1)(0.9) = 0.45$

Continuous Random Variables

Concepts

- PDF $f(x) \geq 0$ with $\int f(x)dx = 1$ over the real line
- $P(a \leq X \leq b) = \int_a^b f(x)dx$; $P(X = c) = 0$ for any single point
- CDF: $F(x) = \int f(t)dt$ up to x, and $f(x) = F'(x)$
- $E[X] = \int x\, f(x)dx$; $Var(X) = E[X^2] - (E[X])^2$
- Common laws: Uniform(a,b), Exponential(λ), Normal(μ,σ^2)
- Exponential: $f(x) = \lambda e^{(-\lambda x)}$, $E = 1/\lambda$, memoryless

Worked example

- Let $f(x) = 2x$ for $0 \leq x \leq 1$, else 0 (e.g. normalised response time)
- Valid: $\int_0^1 2x\, dx = 1$
- $P(X > 0.5) = \int$ from 0.5 to 1 of $2x\, dx = 1 - 0.25 = 0.75$
- $E[X] = \int_0^1 2x^2\, dx = 2/3$
- $E[X^2] = \int_0^1 2x^3\, dx = 1/2$, so $Var(X) = 1/2 - 4/9 = 1/18$

Joint Probability Distribution

Concepts

- Joint PMF: $p(x,y) = P(X=x, Y=y)$; $p \geq 0$ and $\sum\sum p(x,y) = 1$
- Joint PDF (continuous): $f(x,y) \geq 0$, $\iint f(x,y)\,dx\,dy = 1$
- $P((X,Y) \in A) = \sum\sum$ or $\iint$ over A
- $E[g(X,Y)] = \sum\sum g(x,y)\,p(x,y)$
- $Cov(X,Y) = E[XY] - E[X]E[Y]$
- Independent iff $p(x,y) = p_X(x) \cdot p_Y(y)$ for all x, y

Example: X = retransmissions (0,1,2), Y = server state (0 = normal, 1 = overloaded)

p(x,y)	X=0	X=1	X=2
Y=0	0.10	0.20	0.10
Y=1	0.15	0.25	0.20

All entries sum to 1.00
$P(X \geq 1, Y = 1) = 0.25 + 0.20 = 0.45$
$P(X = 0, Y = 0) = 0.10$
$E[X] = 0(0.25)+1(0.45)+2(0.30) = 1.05$

Marginal and Conditional Distributions

Concepts

- Marginal: $p_X(x) = \Sigma_y\, p(x,y)$; $p_Y(y) = \Sigma_x\, p(x,y)$. Continuous: integrate the other variable out
- Conditional: $p(x|y) = p(x,y) / p_Y(y)$, for $p_Y(y) > 0$
- Continuous: $f(x|y) = f(x,y) / f_Y(y)$
- Chain rule: $p(x,y) = p(x|y)\, p_Y(y)$
- Bayes: $p(y|x) = p(x|y)\, p_Y(y) / p_X(x)$

Worked example (same table)

- Marginal of X: 0.25, 0.45, 0.30 (for X = 0, 1, 2)
- Marginal of Y: P(Y=0) = 0.40, P(Y=1) = 0.60
- P(X=1 | Y=1) = 0.25/0.60 = 5/12 ≈ 0.417
- P(Y=1 | X=2) = 0.20/0.30 = 2/3 ≈ 0.667
- Independent? p(1,0) = 0.20 but $p_X(1)p_Y(0)$ = 0.45 × 0.40 = 0.18, so NOT independent: retransmissions depend on server load

Applications in Computer Science

Random & discrete variables

- Generative AI and LLMs (Discrete & Conditional): When a Large Language Model predicts the next word, it uses a discrete random variable over its entire vocabulary. The output is a conditional probability: $P(X_{\text{next}} \mid X_1, X_2, \ldots X_n)$, where the next token is conditionally dependent on the sequence of preceding tokens.
- Networking: packet loss and errors as Binomial; arrivals as Poisson
- Hashing: number of keys per bucket, expected collisions, load factor
- Randomised algorithms: expected running time of quicksort, hashing and skip lists
- Reliability: Geometric law for the number of trials until first success or failure

Continuous variables

- Queueing and performance: exponential inter-arrival and service times (M/M/1)
- Systems: time to failure of disks and servers with Exponential/Weibull models
- Machine learning: Gaussian noise, Gaussian mixtures, weight initialisation
- Simulation: uniform random numbers transformed to other distributions
- Signal and image processing: noise modelled as Normal

Applications in Computer Science

Joint distributions

- ML: joint p(features, label) underlies generative models and Bayesian networks
- NLP: joint probability of word sequences in language models
- Security: joint behaviour of user actions and anomalies in intrusion detection
- Covariance and correlation for feature selection and PCA

Marginal and conditional

- Naive Bayes classifiers: P(class | features) via Bayes rule with conditional independence
- Spam filtering: P(spam | word appears)
- Hidden Markov models and speech recognition: P(state | observation)
- Markov Chains and System State (Conditional): Used in everything from pagerank algorithms to predicting user navigation on a web app. A Markov chain assumes the next state of a random variable depends only on the current state, ignoring the past: $P(X_{t+1} \mid X_t)$
- Marginalisation in probabilistic inference to sum out hidden variables
- Recommender systems: P(user likes item | history)